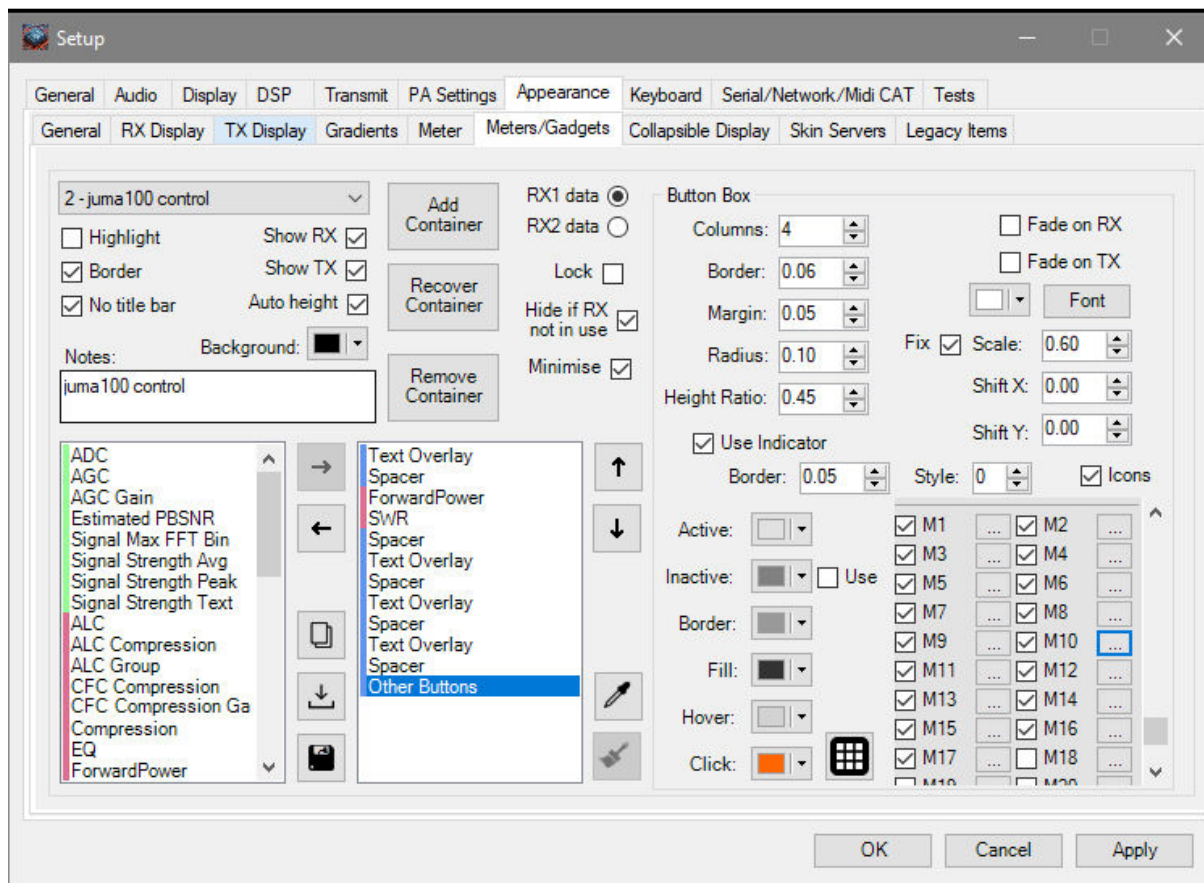


Phil's (VK7ZPD) Notes on Thetis and PA100D band change built on Luca's work (EA8DJR) Adds band change in Thetis with Manual change in the container.

Control of the Juma PA100d is achieved using the server and client applications from Luca's initial instructions and using the same container.

Set up your macro buttons for band change. For clarity I've only shown only M10 the 80m button. If you check the no title bar option the container will be locked in place and the first field isn't blanked when hovering over that area.



Macro Button Configuration - Macro 10

ON state text: Notes:

OFF state text:

☐ Closes parent container
☐ Closes container ☐ Opens container ☐ use closed container for location + size
☐ Closes container ☐ Opens container ☐ use closed container for location + size
☐ Closes container ☐ Opens container ☐ use closed container for location + size
☐ Closes container ☐ Opens container ☐ use closed container for location + size

Button State MMIO Actions CAT Macro

Button State
☐ On
☐ Off
☐ Toggle
☒ Led Indicator 4char
☐ Container visibility
☐ CAT (eg. use [STATE]PS; in macro)
 ON=
☐ Run state cmd when parent container becomes visible

OK Cancel

Set up the MMIO actions as follows. Note that the Mimo message tells the PA100 to switch to 80m via the UDP server, CAT Command tells Thetis/HL2 to switch to 80m band. The cat commands are logical for example 40m is ;ZZBS040; etc The reason for adding the command in both the on and off is to ensure the command is sent on each press.

Macro Button Configuration - Macro 10

ON state text:

Notes:

OFF state text:

☐ Closes parent container

☐ Closes container

☐ Opens container

☐ use closed container for location + size

☐ Closes container

☐ Opens container

☐ use closed container for location + size

☐ Closes container

☐ Opens container

☐ use closed container for location + size

☐ Closes container

☐ Opens container

☐ use closed container for location + size

Button State MMIO Actions CAT Macro

MMIO Actions

	4char	ON raw text message	OFF raw text message
☒ Send message via	xiDM	=B2	=B2
☐ Send message via			
☐ Send message via			
☐ Send message via			

OK

Cancel

Macro Button Configuration - Macro 10

ON state text:
OFF state text:

Notes:

☐ Closes parent container

☐ Closes container
☐ Opens container
☐ use closed container for location + size

☐ Closes container
☐ Opens container
☐ use closed container for location + size

☐ Closes container
☐ Opens container
☐ use closed container for location + size

☐ Closes container
☐ Opens container
☐ use closed container for location + size

Button State

MMIO Actions

CAT Macro

CAT Macro

☒ Run Cat Macro

Cat Tester

?

ZZBS080;

1) ZZBS080;

OK

Cancel

In the CAT Macro tab add the CAT Command

Also added are 3 80m Net frequencies, which change by the same CAT method as the bands. Note that the cat command frequency change string is 11 digits long, in this case the command for 3567 khz is ;ZZFA00003567000;

Macro Button Configuration - Macro 20

ON state text: Notes:

OFF state text:

☐ Closes parent container

☐ Closes container ☐ Opens container ☐ use closed container for location + size

☐ Closes container ☐ Opens container ☐ use closed container for location + size

☐ Closes container ☐ Opens container ☐ use closed container for location + size

☐ Closes container ☐ Opens container ☐ use closed container for location + size

Button State MMIO Actions CAT Macro

CAT Macro

☒ Run Cat Macro 

ZZFA00003640000;

1) ZZFA00003640000;

The finished container will look something like this, I dock mine at the side of Thetis,

JUMA-100D Control |Receive |

0.0W 7.5 15 Power 75 150 0.0W 180+

0.0:1 1.5 SWR 2 3 4 0.0:1 5

Temp:28.0°C |Amp:0.1A|Volt:13.52V

ATTN: 6dB |BAND:80M|FAN:OFF

STATE:OPER|MODE:MANUAL|OK

ALM CLR	OP/SBY	PWR	AUTO
G1 -6dB	G2 -4dB	G3 -2dB	G4 0dB
160m	80m	40m	30m
20m	17m	15m	12m
10m	3567 khz	3580 khz	3640 khz

Clicking on any of the band change buttons should then change the HL2 band and the Juma.

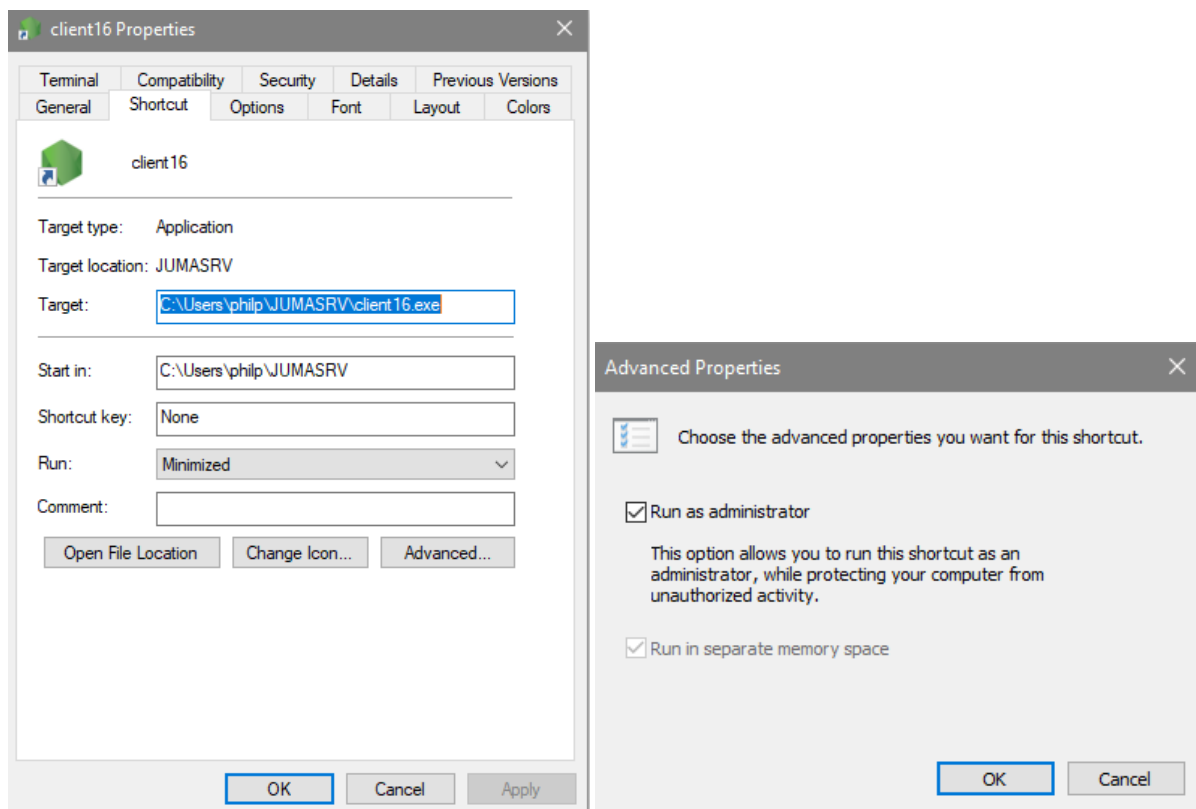
Some additional note. Rather than use the Thetis start options I activate the Server and client via shortcuts and batch files, the reason is that using that method allows those services to be activated as run as administrator and minimised, similarly I close those applications with batch file.

I also set up some container buttons for NR , Setup and Memory as this makes changes quicker than using the Setup menu. Having buttons for NR has an additional advantage that is toggle on and off the desired NR without needing to cycle through the NR options in Thetis.



Batch file start of the server and client are as follows.

Set up shortcuts to the server.batch and client files and flag as Run as Administrator and Run Minimised, the server and client will minimise to the taskbar. The reason for using shortcuts is that batch files can't be run as administrator.



Run both shortcuts with a batch file, in my example below I power on my PSU, run Voicemeeter for audio , and launch Thetis in the first commands.

The start commands launch the server and client with appropriate priviledges and minimised.

The batch file I use.

```
python c:/belfry/power500_switchingon.py
```

```
start c:\belfry\Voicemeeter.lnk
```

```
start c:\belfry\Thetis-HL2.lnk
```

```
start "" "C:\Users\philp\Desktop\Juma Server start\node server.lnk"
```

```
wait 2
```

```
start "" "C:\Users\philp\Desktop\Juma Server start\client16.lnk"
```

to shutdown the server and client I use the following.

```
taskkill /IM client16.exe /F
```

```
taskkill /IM node.exe /F
```

```
wait 10
```

Hope this helps.